

Boundary Conditions at Measurement Extremes

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Abstract

This document applies the Field-Logic framework [1] to measurement regimes.

Reality resolves continuously. Observation samples discretely. Mismatch between resolution rate and sampling rate produces artifacts. Atomic systems and black holes represent the two measurement extremes where sampling-resolution mismatch becomes unavoidable, anchoring the limits examined here.

Undersampling

When the observer sampling rate falls below the system resolution rate, continuous evolution is sampled as discrete states. Intermediate configurations exist but are not recorded. Only endpoints appear. The resulting discontinuity is an artifact of measurement, not a property of the system.

Because the evolution between samples is continuous, outcomes distribute across capture points depending on sampling alignment. This distribution reflects incomplete temporal resolution, not indeterminacy in the underlying process.

The system does not occupy multiple states simultaneously; it occupies one continuous trajectory that is undersampled.

Oversampling

When the observer sampling rate exceeds the system resolution rate, repeated observations capture the same state. Internal evolution occurs below detection threshold. Apparent stasis is an artifact of measurement, not a property of the system.

State changes occur but evolve too slowly to distinguish. Updates connecting internal configurations exist but are not resolved. Only persistence appears. The evolution between effective states is indeterminate to the observer, not to the system.

The system does not cease evolving; its evolution is oversampled.

Measurement Symmetry

Effect	Undersampling	Oversampling
Sampling relation	Too slow	Too fast
Appearance	Discontinuous	Static
Unresolved structure	Between samples	Below detection threshold
Observable	Endpoints only	Persistence only
Dominant artifact	Aliasing, ghosting	Saturation, decay
Underlying evolution	Continuous	Continuous
Boundary	Measurement floor	Measurement ceiling

Same mechanism. Opposite direction. No ontological separation.

Core

A core is a persistent region where phase continuity is locally maximized and closure attempts repeatedly converge. It defines a local reference for phase alignment and persists across all sampling ratios.

A core is not an object, particle, or singularity; it is a persistence condition. A core is identified by the recurrence of closure attempts, not by location or size.

Core strength can be characterized by any conserved inscription density appropriate to the domain.

Distance from Core

Distance from a core does not refer to physical separation. It refers to the phase separation required to maintain bounded resolution.

Circulation requires sufficient phase margin to close without leakage. At zero phase separation, contradiction collapses immediately. Finite phase separation provides the space necessary for circulation to persist.

Shell

A shell is a bounded circulation regime that forms around a core when phase reclosure occurs faster than contradiction leakage.

Shell Formation Criterion

$$\tau_{\text{closure}} < \tau_{\text{leakage}}$$

When this condition holds, a shell forms. When it fails, circulation cannot be sustained.

Measurement Extremes

Atomic systems and black holes represent the two measurement extremes where sampling-resolution mismatch becomes unavoidable. They anchor the limits examined here.

Scale	Closure	Result
Atomic	Fast	Shells stable
Macro	Comparable	Shells lossy
Black hole	Impossible	No shells

Element	Atomic	Macro	Black Hole
Core persistence	Yes	Yes	Yes
Shell formation	Stable	Lossy	Impossible
Closure time	Short	Comparable	Exceeds tolerance
Distance meaning	Hardened	Soft	Fails

Nothing new is introduced across scale. Only boundary conditions change.

This document defines boundary behavior and invariants governing measurement limits.

References

- [1] Jamie L. Thomas. Field-logic: A generative information-theoretic framework for emergent structure and collapse dynamics. Zenodo, 2025. <https://doi.org/10.5281/zenodo.17995347>.